Electromagnetism & Electrodynamics HW1

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Technion

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1 Transformation of co-vectors and contra-vectors

1. Remember that the length element is defined by $\mathrm{d}l^2 = g_{ij}\,\mathrm{d}x^i\,\mathrm{d}x^j$. Using the chain rule, find the transformation rule for g^{ij}.
2. A contra-variant vector v^α is given. Using the transformation rule for contravariant vectors and the metric find the transformation rule for a covector given by

$$v_\alpha = g_{\alpha\beta} v^\beta \tag{1}$$

3. show that if $w_\alpha = g_{\alpha\beta} u^\beta$ and $v^\alpha = g^{\alpha\beta} w_\beta$ then $u^\alpha = v^\alpha$

1.1 Metric transformation

Directly

$$\begin{aligned}
\mathrm{d}l^2 &= g_{ij}\,\mathrm{d}x^i\,\mathrm{d}x^j && (2)\\
&= g_{ij}\frac{\mathrm{d}x^i}{\mathrm{d}\left(x'\right)^m}\mathrm{d}\left(x'\right)^m \frac{\mathrm{d}x^j}{\mathrm{d}\left(x'\right)^n}\mathrm{d}\left(x'\right)^n && (3)\\
&= \left(\Lambda^{-1}\right)^i{}_m g_{ij}\left(\Lambda^{-1}\right)^j{}_n \mathrm{d}\left(x'\right)^m \mathrm{d}\left(x'\right)^n && (4)
\end{aligned}$$

Therefore

$$g_{ij} \to \left(\Lambda^{-1}\right)^i{}_m g_{ij}\left(\Lambda^{-1}\right)^j{}_n \tag{5}$$

1.2 contra to co vector

$$\begin{aligned}
v_\alpha \to v'_\gamma &= g'{}_{\gamma\delta} v'^\delta && (6)\\
&= \frac{\mathrm{d}x^\alpha}{\mathrm{d}x'^\gamma}\frac{\mathrm{d}x^\beta}{\mathrm{d}x'^\delta} g_{\alpha\beta}\frac{\mathrm{d}x'^\delta}{\mathrm{d}x'^\zeta} v^\zeta && (7)\\
&= \frac{\mathrm{d}x^\alpha}{\mathrm{d}x'^\gamma} g_{\alpha\beta}\delta^\beta{}_\zeta v^\zeta && (8)\\
&= \frac{\mathrm{d}x^\alpha}{\mathrm{d}x'^\gamma} g_{\alpha\beta} v^\beta && (9)\\
&= \frac{\mathrm{d}x^\alpha}{\mathrm{d}x'^\gamma} v_\alpha && (10)\\
&= \left(\Lambda^{-1}\right)^\alpha{}_\gamma v_\alpha && (11)
\end{aligned}$$

1.3 show equality

$$\begin{aligned}
v^\alpha &= g^{\alpha\beta} w_\beta && (12)\\
&= g^{\alpha\beta} g_{\beta\gamma} u^\gamma && (13)\\
&= g^{\alpha\beta} u_\beta && (14)\\
&= u^\alpha && (15)
\end{aligned}$$

2 Transformation of 2D minkowski space

The conserved length in 2D Minkowski space is

$$\mathrm{d}l^2 = g^{ij}\,\mathrm{d}x_i\,\mathrm{d}x_j \tag{16}$$

for

$$g^{ij} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{17}$$

we'll look at the coordinate transformations

$$x' = ax + by \qquad\qquad y' = cx + dy \tag{18}$$

find

$$\Lambda = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \tag{19}$$

for which the matrix doesn't change, i.e

$$g = \Lambda^T g \Lambda \tag{20}$$

notice the transformation is an isometry of the metric.

Performing the matrix multiplication

$$\Lambda^T g \Lambda = \begin{pmatrix} a & c \\ b & d \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \tag{21}$$

$$= \begin{pmatrix} a & c \\ b & d \end{pmatrix} \begin{pmatrix} a & b \\ -c & -d \end{pmatrix} \tag{22}$$

$$= \begin{pmatrix} a^2 - c^2 & ab - cd \\ ab - cd & b^2 - d^2 \end{pmatrix} \tag{23}$$

we must have

$$ab - cd = 0 \qquad a^2 - c^2 = 1 \qquad b^2 - d^2 = -1 \tag{24}$$

if we use the hint to define

$$a = \cosh\varphi \qquad c = \sinh\varphi \qquad d = \cosh\varphi \qquad b = \sinh\varphi \tag{25}$$

then

$$a^2 - c^2 = \cosh^2\varphi - \sinh^2\varphi = 1 \tag{26}$$

$$b^2 - d^2 = \sinh^2\varphi - \cosh^2\varphi = -1 \tag{27}$$

$$ab - cd = \cosh\varphi\sinh\varphi - \sinh\varphi\cosh\varphi = 0 \tag{28}$$

$$ac - bd = \cdots = 0 \tag{29}$$

giving us what's needed, so

$$\Lambda = \begin{pmatrix} \cosh\varphi & \sinh\varphi \\ \sinh\varphi & \cosh\varphi \end{pmatrix} \tag{30}$$

3 flat and no very flat metrics

1. How many degrees of freedom does the metric have in d dimensions? How many degrees of freedom does a coordinate transofmation have in d dimensions?
2. We'll look at an infinite cylinder of radius R. We'll mark its surface with S. S is a two dimensional space and can be described by a shape in three dimensional euclidian space:

$$x^2 + y^2 = R^2 \tag{31}$$

   Move to cylindrical coordinates and find the metric for S. Find a coordinate system where the metric is euclidian, i.e $g_{ij} = \delta_{ij}$

3. Calculate the metric on the surface of a sphere with radius R.

3.1 degrees of freedom

The metric has $\frac{d(d+1)}{2}$ degrees of freedom in d dimensions. This is because the metric is a symmetric tensor. The coordinate transformation has d degrees of freedom because there are d new coordinate functions.

3.2 Cylinder

We'll move to cylindrical corodinate such that

$$x = R\cos\theta \qquad y = R\sin\theta \qquad z = z \tag{32}$$

Therefore

$$\mathrm{d}x = -R\sin\theta\,\mathrm{d}\theta \qquad \mathrm{d}y = R\cos\theta\,\mathrm{d}\theta \qquad \mathrm{d}z = \mathrm{d}z \tag{33}$$

giving us

$$\mathrm{d}l^2 = R^2\,\mathrm{d}\theta^2 + \mathrm{d}z^2 \tag{34}$$

so the metric is

$$g_{ij} = \begin{pmatrix} R^2 & 0 \\ 0 & 1 \end{pmatrix} \tag{35}$$

if we move to a coordinate system such that

$$u = R\theta \tag{36}$$

we get

$$\mathrm{d}l^2 = \mathrm{d}u^2 + \mathrm{d}z^2 \tag{37}$$

or in other words

$$g_{ij} = \delta_{ij} \tag{38}$$

3.3 surface of sphere

In spherical coordinates the surface is given by

$$x = R\sin\theta\cos\varphi \qquad y = R\sin\theta\sin\varphi \qquad z = R\cos\theta \tag{39}$$

the line element in spherical coordinates is given by

$$\mathrm{d}l^2 = \mathrm{d}r^2 + r^2\,\mathrm{d}\theta^2 + r^2\sin^2\theta\,\mathrm{d}\varphi^2 \tag{40}$$

Since $r = R$ is constant $\mathrm{d}r = 0$ so

$$\mathrm{d}l^2 = R^2\,\mathrm{d}\theta^2 + R^2\sin^2\theta\,\mathrm{d}\varphi^2 \tag{41}$$

so the metric is

$$g_{ij} = \begin{pmatrix} R^2 & 0 \\ 0 & R^2\sin^2\theta \end{pmatrix} \tag{42}$$